Tutorial 9
Columns

---

1. Determine the maximum force P that can be applied to the handle so that the $A-36$ steel control rod BC does not buckle (see Fig. 1). The rod has a diameter of 25 mm and $E = 200$ GPa.

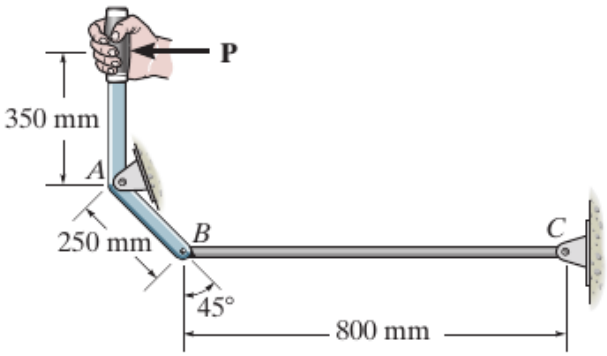


Fig. 1

2. Determine the maximum allowable intensity w of the distributed load that can be applied to member BC without causing member AB to buckle (see Fig. 2). Assume that AB is made of steel and is pinned at its ends for $x-x$ axis buckling and fixed at its ends for $y-y$ axis buckling. Use a factor of safety with respect to buckling of 3. $E_{st} = 200\,\text{GPa}$, $\sigma_Y = 360\,\text{MPa}$.

3. The column with constant EI has the end constraints shown in Fig. 3. Determine the critical load for the column.

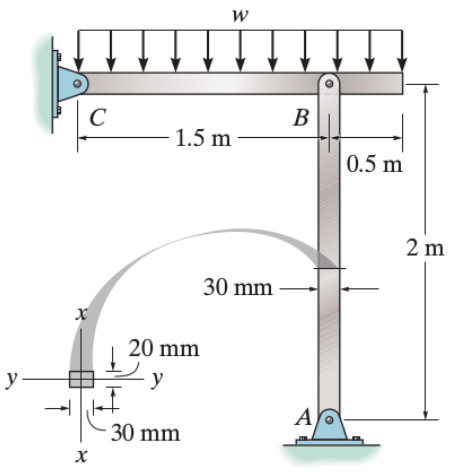


Fig. 2

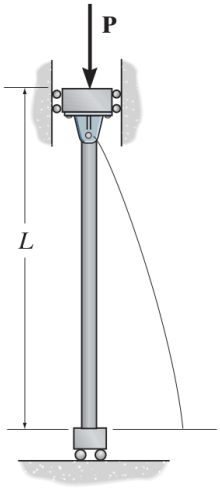


Fig. 3

4. The depicted beam (see Fig. 4)is subjected to axial compression and is supported at both ends elastically by torsional springs. Determine the critical load.

   Given:

$$EI = lc_T.$$

5. The depicted beam (see Fig. 5) consists of a rigid and an elastic part (flexural rigidity EI). Determine the bucking condition and the critical load.

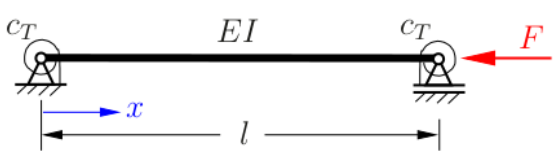


Fig. 4

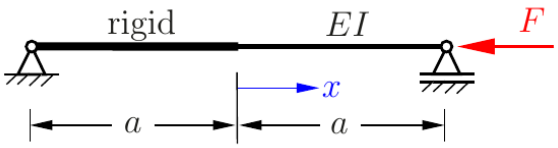


Fig. 5